

Standard Operating Procedure: Industrial Equipment Inspection

1. PURPOSE

This document establishes the standard operating procedure for the inspection of industrial equipment at the manufacturing facility. All personnel involved in equipment maintenance and quality assurance must follow these guidelines to ensure workplace safety, regulatory compliance, and operational reliability.

2. SCOPE

This SOP applies to all rotating machinery, pressure vessels, electrical panels, and conveyor systems within Plant A and Plant B. It covers routine visual inspections, scheduled preventive maintenance checks, and unscheduled breakdown investigations.

3. RESPONSIBILITIES

The Plant Manager is responsible for ensuring that all inspections are carried out on schedule. The Maintenance Supervisor assigns inspection tasks to qualified technicians. Each technician must hold a valid certification for the equipment category they inspect. The Quality Assurance department reviews inspection reports within 48 hours.

4. INSPECTION PROCEDURE

4.1 Pre-Inspection Preparation

Before beginning any inspection, the technician must: (a) review the equipment maintenance history from the CMMS database, (b) gather all required tools and personal protective equipment, (c) obtain a valid work permit if the equipment is in a hazardous zone, (d) notify the control room operator that the equipment will be taken offline.

4.2 Visual Inspection

Examine the external surfaces for corrosion, cracks, leaks, or unusual discoloration. Check all fasteners for tightness. Verify that safety guards and covers are properly secured. Inspect wiring and cable trays for damage or overheating. Document findings with photographs.

4.3 Functional Testing

After visual inspection, perform functional tests as specified in the equipment manual. Record vibration levels using a portable analyzer. Measure operating temperature with an infrared thermometer. Compare readings against baseline values from the equipment commissioning report.

5. REPORTING AND DOCUMENTATION

All inspection findings must be recorded in the Inspection Report Form (IRF-2024). The report must include: date and time of inspection, equipment ID and location, inspector name and certification number, detailed findings with severity classification (Critical, Major, Minor), recommended corrective actions, and a follow-up schedule.

6. SAFETY REQUIREMENTS

Lockout/Tagout (LOTO) procedures must be followed before inspecting any energized equipment. Confined space entry permits are required for internal vessel inspections. Fall protection equipment must be used when working at heights above 1.8 meters. Emergency procedures must be reviewed before entering any hazardous area.

7. REVISION HISTORY

Rev 1.0 ? Initial release, January 2024

Rev 1.1 ? Added confined space entry requirements, March 2024

Rev 1.2 ? Updated vibration analysis thresholds, July 2024